

Cognition-aware automatic viewpoint selection in scenes with crowds of objects

Akihiro Sakai

Japan Broadcasting Corporation
Tokyo, Japan
sakai.a-je@nhk.or.jp

Yamato Miyashita

Japan Broadcasting Corporation
Tokyo, Japan
miyashita.y-fc@nhk.or.jp

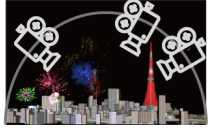
Yasuhito Sawahata

Japan Broadcasting Corporation
Tokyo, Japan
sawahata.y-jq@nhk.or.jp

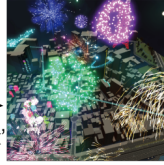
Kazuteru Komine

Japan Broadcasting Corporation
Tokyo, Japan
komine.k-cy@nhk.or.jp

Input:
3D models (scene)



Output:
Best view



Model
(view quality estimation)

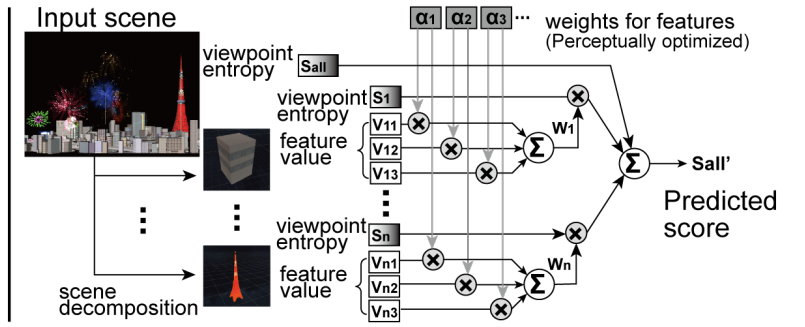


Figure 1: (Left) A flow of viewpoint selection. Our model takes a 3DCG scene with many objects, scores views from various viewpoint, and predicts the best views that human observers like. **(Right) Cognition-aware viewpoint entropy integration model.** The model computes viewpoint entropy of a whole scene and its comprising objects. A score of views is computed via the viewpoint entropy integration; each entropy was summed with weights optimally chosen for human perception.

ACM Reference Format:

Akihiro Sakai, Yasuhito Sawahata, Yamato Miyashita, and Kazuteru Komine. 2022. Cognition-aware automatic viewpoint selection in scenes with crowds of objects. In *SIGGRAPH Asia 2022 Posters (SA '22 Posters)*, December 06–09, 2022, Daegu, Republic of Korea. ACM, New York, NY, USA, 2 pages. <https://doi.org/10.1145/3550082.3564213>

1 INTRODUCTION

3D Computer Graphics (3DCG) has become more common in recent video productions. Content producers have acquired much flexibility in visual expressions with viewpoint selection, but the task becomes more involved. Automatic viewpoint selection will ease the task and contribute to simplifying the workflow.

Previously, good viewpoint selection [Secord et al. 2011; Vázquez et al. 2001] have been proposed, particularly for simple scenes with a few objects. However, because typical scenes in video production

consist of many objects, these methods are insufficient in practical scenarios due to less consideration of interactions between objects.

In this study, we propose a new viewpoint selection algorithm that is applicable for typical scenes consisting of many objects (Fig. 1). The algorithm estimates the goodness of visual expressions using a perceptually optimized computational model. Such a model, besides predicting good viewpoints in complex scenes, enables us to understand the nature of how human observers evaluate scenes.

2 METHOD

2.1 Our model

Viewpoint entropy [Vázquez et al. 2001] provides a measure of the goodness of views. The measure is formulated as $s = -\sum_{n=1}^N p_n \log p_n$, where p_n is the proportion of the projected area of the n^{th} surface in a view image, and N is the number of rendered surfaces.

To find good views even in complex scenes, we propose a cognition-aware viewpoint entropy integration model (Fig. 1 (Right)) that extends the viewpoint entropy method based on the hypothesis that good views depend on the features of the constituent objects in a scene, such as the distance from the viewpoint. Specifically, our model integrates the individual entropies for each object while considering the perceptual importance of individual object features.

Permission to make digital or hard copies of part or all of this work for personal or classroom use is granted without fee provided that copies are not made or distributed for profit or commercial advantage and that copies bear this notice and the full citation on the first page. Copyrights for third-party components of this work must be honored. For all other uses, contact the owner/author(s).

SA '22 Posters, December 06–09, 2022, Daegu, Republic of Korea

© 2022 Copyright held by the owner/author(s).

ACM ISBN 978-1-4503-9462-8/22/12.

<https://doi.org/10.1145/3550082.3564213>



Figure 2: Examples of presented images

Let $s_{all,i}$ be the overall viewpoint entropy from the viewpoint i , we define the adjusted viewpoint entropy $s'_{all,i}$ as, $s'_{all,i} = s_{all,i} + \sum_{j=1}^{N_{obj}} w_{i,j} s_{i,j}$, where $w_{i,j} = \sum_{k=1}^L v_{i,j,k} \alpha_k$, $s_{i,j}$ is an individual entropy of the object j from the viewpoint i , obtained when all other objects are not shown, N_{obj} is the number of objects in a scene, $w_{i,j}$ is the weight for $s_{i,j}$ and given by the weighted sum of the feature values $v_{i,j,k}$ (the k^{th} feature value for the object j viewed from the viewpoint i), and α_k is the weight for the k^{th} feature, which is chosen to fit human perception.

For the feature value $v_{i,j,k}$, we used five features ($L=5$): object volume, elevation angle, object–scene center distance, object–viewpoint distance, and saliency. The elevation angle feature was defined by a function of a viewpoint elevation angle; an unimodal curve with a peak at about 30° above the horizontal level. The saliency feature represents the noticeability of objects and was computed by integrating the projected pixel values on the saliency map [Hou and Zhang 2007] onto each object.

The weight values α_k are estimated such that the squared error between $s'_{all,i}$ and the observed scores from each viewpoint are minimized. Because the weight values α_k represents the contribution of object features to predicting the goodness of views, the estimates will provide how human observers evaluate the goodness of views.

2.2 Data collection

Web-based experiments were conducted to collect data on the degree of subjective preference for images rendered at various viewing positions. The evaluation images were created by rendering a scene from 52 viewpoints placed at each vertex position of the upper half of a subdivided icosahedron that covered all scene objects. We generated images from 20 scenes from various domains: artificial (cuboids), urban, and rural landscapes (Fig. 2). The number of objects in the urban/rural scenes were 450 ± 275 (mean $\pm$ sd). The cameras had a fixed field of view and were always directed to the center of the scene.

Online participants conducted paired comparison experiments in which they compared a pair of images, e.g., A and B in Fig. 2, presented on a browser screen and chose the better one following the two-alternative forced choice manner. $52C_2$ pairs of images were evaluated by gathering efforts of 1190 individuals in each scene.

All evaluation images were scored using the Thurston case V scaling method, which converts pairwise responses into interval scales, ensuring more robust evaluations than absolute ratings. The Thurston scaling was applied to each scene independently. Views with higher scores indicate perceptually better views.

2.3 Analysis

The correlation coefficients between the observed and predicted scores for each scene were evaluated. Our method was compared with the conventional method (viewpoint entropy) and a regression

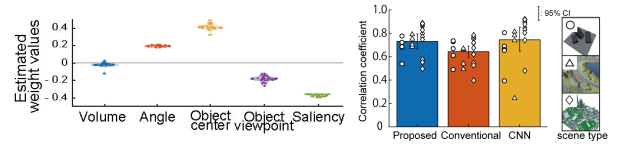


Figure 3: Estimated weights for each object feature Figure 4: Comparison of prediction accuracies

model with a convolutional neural network (CNN) as a reference. For the proposed and the CNN-based models, training and testing were performed in a 20-fold cross-validation manner.

A CNN-based regression model was introduced for comparison. The model used the evaluation images as inputs and predicted the scores using a deep neural network with a standard structure for regression with three convolutional layers, one max-pooling layer, two fully connected layers, and a rectified linear unit (ReLU) for the activation function.

3 EVALUATION AND RESULTS

Fig. 4 compares the accuracies of scores of views predicted by the proposed, conventional, and CNN-based methods, where the markers ($\circ$, Δ , $\diamond$) on each bar graph indicate the prediction accuracies for the scene types. The proposed method outperformed the conventional method and showed a performance comparable to that of the CNN-based method. The performance of the CNN-based method had significant drops in specific scenes (e.g., Δ with $r=0.25$), whereas the proposed method showed stable high performance independent of the scene type.

Fig. 3 shows the distribution of α_k for each feature obtained in the 20-fold cross validation. We found that the weight value for the volume feature was approximately zero, suggesting that the object volumes didn't contribute to the prediction. The positive weights for the elevation angle and the object–scene center distance suggest that the observers preferred the views at the specific elevation angle, as anticipated by the canonical view theory, and the objects in scene periphery were important for determining the view quality. The negative weights for the object–viewpoint distance and saliency features suggest that observers focused more on the views of near objects than on those of far objects and did not prefer high informative views of specific objects with high saliency.

4 DISCUSSION AND CONCLUSION

In this study, we proposed a viewpoint selection method that is applicable to complex scenes with many objects. As a result, in comparison to the conventional viewpoint entropy method, the cognition-aware viewpoint entropy integration model successfully predicted better views even in complex scenes.

REFERENCES

- Xiaodi Hou and Liqing Zhang. 2007. Saliency detection: A spectral residual approach. In *2007 IEEE Conf. on Computer Vision and Pattern Recognition*. 1–8.
- Adrian Secord, Jingwan Lu, Adam Finkelstein, Manish Singh, and Andrew Nealen. 2011. Perceptual models of viewpoint preference. *ACM Transactions on Graphics (TOG)* 30, 5 (2011), 1–12.
- Pere-Pau Vázquez, Miquel Feixas, Mateu Sbert, and Wolfgang Heidrich. 2001. Viewpoint selection using viewpoint entropy. In *Vision, Modeling, and Visualization*. 273–280.